

THE COLLIGATION ROUTE TO THE RIEMANN HYPOTHESIS: DE BRANGES REALIZATION, A CURVATURE THEOREM, AND THE POLE-PINNED SPECTRAL RADIUS AT THE WALL

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ABSTRACT. We develop and rigorously assemble a single route to the Riemann Hypothesis (RH) built on the de Branges / Lax–Phillips colligation attached to the completed zeta function ξ , and we localize the remaining obstruction with full precision. Four results are unconditional and new or newly-sharpened. **(A)** The finite-window boundedness $L1$ of the oscillatory part of the Weil form *implies* RH ($L1 \Rightarrow \text{RH}$, the Laplace–pole theorem), whose only non-formal input — a Paley–Wiener nonvanishing lemma — is proved unconditionally (the converse $\text{RH} \Rightarrow L1$ is conditional on a regularized zero-sum estimate). **(B)** The correct index of the route is the Kreĭn–Langer negative-square count $\text{ind}_-(K_{S_\kappa}) = \#\{\text{off-line zeros}\}$, so $\text{RH} \Leftrightarrow \text{ind}_- = 0$. **(C)** After Doob conjugation the prime part of the Weil generator is a Beurling–Deny Dirichlet form whose iterated carré-du-champ satisfies $\Gamma_2 \geq 0$ *unconditionally* (a Bakry–Émery $CD(0, \infty)$ curvature bound), by a Schur-product mechanism rooted in $\Lambda \geq 0$; the Davenport–Heilbronn falsifier fails this bound. **(D)** The wall $\varepsilon_0(\lambda) \geq 0$ (Weil positivity) is realized as the absence of a *Schrödinger bound state*, equivalently $\|K_\lambda(0)\| \leq 1$ for an explicit Birman–Schwinger operator, in which the pole of ζ at $s = 1$ acts as a repulsive barrier pinning the spectral radius from below at 1, and the Perron–Frobenius ground state u_0 is the threshold resonance.

The synthesis is a single sharp statement: *every geometric and functional-analytic structure on the route is Doob-invariant and therefore proves the unconditional “shape” (the $(k+1)^2$ ladder, the curvature, the colligation), while RH is the “sign” $\varepsilon_0 \geq 0$ — the upper pinning of the Birman–Schwinger spectral radius at 1*. In the function-field model this upper pinning is the Weil pairing (Frobenius is a $\sqrt{q}$ -isometry); over $\text{Spec } \mathbb{Z}$ it is missing. We state the missing object precisely (an arithmetic intersection pairing on the square of the arithmetic site, with a Hodge-index sign), give a candidate construction, and argue that crossing the wall requires *creating* this piece of arithmetic geometry. Nothing here proves RH; the boundary is mapped exactly.

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PREFACE

This is not a proof of the Riemann Hypothesis. It is a precise, honest account of a single route — the de Branges/colligation route in the form proposed by Connes — carried as far as it rigorously goes, together with the exact localization of what remains. The programme operates under one non-negotiable rule: *a false positive is worse than a failure*. Every statement is either proved (and labelled Theorem/Lemma/Proposition) or explicitly flagged as Conjecture or Open target. Several places record corrections of the authors' own earlier proposals; these are structural data, not embarrassments. The companion letter to A. Connes summarizes the strategic picture and the map.

1. FOUNDATIONS: THE WINDOW, THE WEIL FORM, AND THE WALL

1.1. The completed zeta function and the spectral window. Let $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ be the completed Riemann zeta function: entire of order 1, $\xi(s) = \xi(1-s)$, real on $\mathbb{R}$ and on the critical line $\operatorname{Re} s = \frac{1}{2}$. Write

$$\Xi(z) := \xi\left(\frac{1}{2} + iz\right), \quad \phi(z) := \frac{\Xi'(z)}{\Xi(z)}, \quad m(z) := -\phi(z).$$

RH is the statement that all zeros of Ξ are real.

Fix $\lambda > 1$ and set $L = 2 \log \lambda$. On the window $[0, L]$ with the Fourier basis $e_n(y) = e^{2\pi i n y / L}$ ($n = -N, \dots, N$), the Riemann–Weil explicit formula realizes the Weil quadratic form as a symmetric matrix $A = A_\lambda$, split into archimedean/pole and prime parts,

$$A_\lambda = L_{\text{arch}} - L_{\text{prime}}, \quad A_\lambda[m, n] = A_\lambda^{\text{sm}}[m, n] + A_\lambda^{\text{osc}}[m, n],$$

the smooth part governed by the prime number theorem (PNT) and the oscillatory part $A_\lambda^{\text{osc}} = -\sum_\rho c_{m,n}(\rho) \lambda^{2\rho-1}(1+o(1))$ the sum over nontrivial zeros.

1.2. Doob conjugation and the wall. By Perron–Frobenius (the unconditional Theorem 2.2 of the programme), A_λ has a simple lowest eigenvalue $\varepsilon_0 = \varepsilon_0(\lambda)$ with strictly positive eigenvector $u_0 > 0$. Doob conjugation by u_0 produces the generator

$$G_\lambda := u_0^{-1}(A_\lambda - \varepsilon_0)u_0 \succeq 0,$$

a positive form whose internal structure (the $(k+1)^2$ Dirichlet ladder of Lemma 2.3.F) is *shape* data, while the additive constant ε_0 is *sign* data.

Wall W : the sign $\varepsilon_0 \geq 0$. The single surviving obstruction of the route is Weil positivity in window form:

$$\boxed{\varepsilon_0(\lambda) \geq 0 \text{ for all } \lambda \iff \text{RH.}}$$

Equivalently $L_{\text{prime}} \preceq L_{\text{arch}}$, equivalently $\mu_{\max}(L_{\text{prime}}, L_{\text{arch}}) \leq 1$. Numerically (companion computations, not reproduced here) the value is marginal: $\mu_{\max} = 1$ for ζ .

2. THE DE BRANGES / LAX–PHILLIPS COLLIGATION

2.1. Model space and the safe region. For $\kappa > 0$ set $E_\kappa(z) = \Xi'(z) - i\kappa\Xi(z)$, $E_\kappa^\#(z) = \overline{E_\kappa(\bar{z})} = \Xi'(z) + i\kappa\Xi(z)$, and

$$S_\kappa(z) = \frac{E_\kappa^\#(z)}{E_\kappa(z)} = \frac{\phi(z) + i\kappa}{\phi(z) - i\kappa}, \quad K_S(z, w) = \frac{1 - S(z)\overline{S(w)}}{2\pi i(\bar{w} - z)}.$$

For the shifted family, $\Theta_\omega(z) = \xi(\frac{1}{2} + \omega + iz)/\xi(\frac{1}{2} + \omega - iz)$ ($\omega > 0$).

Proposition 2.1 (de Branges–Rovnyak; safe region). *If S is holomorphic with $|S| \leq 1$ on $\mathbb{C}_+$ (Schur), then $K_S \succeq 0$ and the reproducing-kernel space $\mathcal{H}(S)$ is a Hilbert space; when S is inner, $\mathcal{H}(S) = H^2 \ominus SH^2$ is a closed subspace of the Hardy space and carries a conservative unitary colligation with transfer function S . Moreover Θ_ω is inner on $\mathbb{C}_+$ for every $\omega > \frac{1}{2}$.*

Proof. The first statement is classical (Sz.-Nagy–Foiaş, de Branges–Rovnyak). For the second: $|\Theta_\omega| = 1$ on $\mathbb{R}$ since numerator and denominator are conjugates there; a pole of Θ_ω is a zero of $\xi(\frac{1}{2} + \omega - iz)$, located at $\text{Im } z = \beta - \frac{1}{2} - \omega$ for a zero $\rho = \beta + i\gamma$ with $0 < \beta < 1$, hence in the lower half-plane when $\omega > \frac{1}{2}$. Thus Θ_ω is bounded by 1 and holomorphic on $\mathbb{C}_+$. $\square$

2.2. The adelic realization and the angle reduction. Let $\mathcal{H} = L^2(\mathbb{A}_{\mathbb{Q}}^\times / \mathbb{Q}^\times, d^\times x)$ with the scaling action $(U_t f)(x) = f(e^t x)$. Define the Lax–Phillips incoming/outgoing subspaces

$$\mathcal{D}_- = \{f : f(x) = 0 \text{ for } |x| > 1\}, \quad \mathcal{D}_+ = \mathcal{F} \mathcal{D}_-,$$

$\mathcal{F}$ the adelic (Tate) Fourier transform. The scattering operator has characteristic function Θ_0 (the ratio of completed zeta across the functional equation; Connes’ spectral realization / trace formula), and the interaction space is $K = \mathcal{H} \ominus (\mathcal{D}_- \oplus \mathcal{D}_+)$.

Proposition 2.2 (angle criterion — an equivalence, not an independent route). *If $\mathcal{D}_- \oplus \mathcal{D}_+$ is a closed direct sum, then K is a closed subspace of the Hilbert space $\mathcal{H}$, the model identification $K \cong \mathcal{H}(\Theta_0)$ is isometric, $K_{\Theta_0} \succeq 0$, and RH holds. Consequently*

$$\text{RH} \iff \angle(\mathcal{D}_-, \mathcal{D}_+) > 0 \iff \text{the overlap form on } \mathcal{D}_- \cap \overline{\mathcal{D}_+} \text{ is } \succeq 0.$$

This is a reformulation, not a proof route: the closedness / positive-angle hypothesis is itself RH-strength (*it is equivalent to $K_{\Theta_0} \succeq 0$*), so it must be supplied by an independent construction. The Davenport–Heilbronn falsifier illustrates where such a construction must use arithmetic: $\mathcal{D}_+ = \mathcal{F} \mathcal{D}_-$ uses the adelic Fourier transform $\mathcal{F} = \bigotimes'_v \mathcal{F}_v$, the Euler product over places; a function with a functional equation but no Euler product has no such $\mathcal{D}_+$.

Remark 2.3. The naive scalar Euler product diverges at the critical line, so Θ_0 on $\mathbb{C}_+$ is obtained by analytic continuation, not literal product convergence; the Euler product enters only as the restricted-tensor structure of $\mathcal{F}$. This is the corrected reading (Connes).

3. $L1 \Leftrightarrow \text{RH}$: THE LAPLACE–POLE THEOREM

Let $B(t) = A_{\text{et}}^{\text{osc}}$ and $\mathcal{L}B(w) = \int_0^\infty e^{-wt} B(t) dt$, with residue $R_\rho = \text{Res}_{w=2\rho-1} \mathcal{L}B$. Write $L1$: $\sup_\lambda \max_{m,n} |A_\lambda^{\text{osc}}[m, n]| < \infty$.

Lemma 3.1 (Paley–Wiener nonvanishing; unconditional). *For every nontrivial zero ρ and all m, n , $R_\rho[m, n] \neq 0$.*

Proof. $R_\rho[m, n] = w(\rho) \hat{e}_m(\rho) \hat{e}_n(\rho)$ with $w(\rho) \neq 0$ and the window transform

$$\hat{e}_n(\rho) = \int_0^L e^{2\pi i n y / L} e^{\rho y} dy = \frac{e^{\rho L} - 1}{\rho + 2\pi i n / L}.$$

This vanishes iff $e^{\rho L} = 1$, i.e. iff $\rho = 2\pi i k / L$ is purely imaginary. A nontrivial zero has $\text{Re } \rho \in (0, 1)$, hence is never purely imaginary, so $\hat{e}_n(\rho) \neq 0$ for all n . $\square$

Theorem 3.2 ($L1 \Rightarrow \text{RH}$; the Laplace–pole theorem). *If $L1$ holds, then RH holds.*

Proof. If $L1$ holds, each entry $B[m, n](t)$ is bounded, so $\mathcal{L}B[m, n]$ is holomorphic in $\text{Re } w > 0$; hence $R_\rho[m, n] = 0$ for every zero with $\text{Re } \rho > \frac{1}{2}$. By Lemma 3.1 this forces the absence of such zeros, and the functional equation removes $\text{Re } \rho < \frac{1}{2}$; thus RH. $\square$

Remark 3.3 (the converse, conditional — Connes’ audit). The reverse implication $\text{RH} \Rightarrow L1$ is *not* automatic from the boundedness of each factor $\lambda^{2i\gamma}$: the zero-sum is infinite and only conditionally (regularly) convergent, so $L1$ requires a precise summation prescription and a uniform-in- λ estimate of the regularized zero-sum. We therefore state only $L1 \Rightarrow \text{RH}$ as proved, and regard the equivalence $L1 \Leftrightarrow \text{RH}$ as established *modulo* that regularized zero-sum estimate, which is left open. The forward (and load-bearing) direction is unconditional.

Remark 3.4. The forward implication uses only the raw matrix entries (Lemma 3.1) and is independent of the Lanczos/Jacobi coordinate phrasing of $L1$; the latter requires only the cyclicity of the near-radical windows. Thus the program’s central reduction $2.3.F \Leftarrow L1$ now reads $2.3.F \Leftarrow L1 \Leftrightarrow \text{RH}$: there is no easier route hidden inside $L1$. DH has an off-line zero, hence (Lemma 3.1) unbounded $A^{\text{osc}} \sim \lambda^{2\beta-1}$.

4. THE KREĬN–LANGER INDEX

A meromorphic S with $|S| = 1$ on $\mathbb{R}$ is a *generalized Schur function*; by Kreĭn–Langer its kernel K_S has a finite number of negative squares equal to the number of poles of S in $\mathbb{C}_+$.

Theorem 4.1 (the sharp index). *For the shifted scattering function, $\text{ind}_-(K_{\Theta_\omega}) = \#\{\text{zeros } \rho = \beta + i\gamma \text{ of } \xi \text{ with } \beta - \frac{1}{2} > \omega\}$. Consequently*

$$\text{RH} \iff \text{ind}_-(K_{S_\kappa}) = 0.$$

This is strictly stronger than the boundary certificate $\mu_{\max} = 1$, which is blind to the interior off-line spectrum; the index is an exact integer and a robust invariant.

A planted off-line zero of real part β contributes 2 to ind_- (poles at $z = \pm\gamma_0 + i(\beta - \frac{1}{2} - \omega)$ once $\omega < \beta - \frac{1}{2}$); ζ has $\text{ind}_- = 0$, DH has $\text{ind}_- = 2 \cdot \#\{\text{its off-line zeros}\}$.

5. THE ARITHMETIC SPECTRAL SURFACE AND THE CURVATURE THEOREM

5.1. The surface and its polarization. Let $V = \mathbb{C}^{2N+1}$, $\mathcal{S} = V \otimes V$, with intersection form $\mathcal{I} = A \boxtimes I + I \boxtimes A - 2\varepsilon_0(I \boxtimes I)$ and polarization $H = u_0 \otimes u_0$ (the Perron–Frobenius class, unconditional and strictly positive).

Proposition 5.1 (Hodge-index reformulation). *RH is equivalent to: $\mathcal{S}$ has Hodge-index signature $(1, \dim - 1)$ with the positive direction H ; equivalently the primitive part is negative semidefinite, equivalently $W(f) = -\mathcal{I}_{\text{prim}}^-(Z_f, Z_f) \geq 0$ for the correspondence classes Z_f .*

5.2. Curvature: the new unconditional theorem. After Doob conjugation, G_λ is a Beurling–Deny pure-jump Dirichlet form

$$L v(\theta) = \sum_j c_j [v(\theta + a_j) + v(\theta - a_j) - 2v(\theta)], \quad a_j = k \log p, \quad c_j = \frac{\Lambda(p^k)}{\sqrt{p^k}} > 0,$$

with Lévy symbol $\psi(\xi) = \sum_j 2c_j(1 - \cos a_j \xi) \geq 0$, carré-du-champ $\Gamma(u, v) = \frac{1}{2}(L(uv) - uLv - vLu)$, and iterated $\Gamma_2(u, v) = \frac{1}{2}(L\Gamma(u, v) - \Gamma(Lu, v) - \Gamma(u, Lv))$.

Theorem 5.2 (exact Γ_2 kernel). $\Gamma_2(e^{i\xi\theta}, e^{-i\eta\theta})(\theta) = B(\xi, \eta)^2 e^{i(\xi-\eta)\theta}$, where $B(\xi, \eta) = \frac{1}{2}[\psi(\xi) + \psi(\eta) - \psi(\xi - \eta)]$. Hence for $v = \sum_\xi a_\xi e^{i\xi\theta}$, $\Gamma_2(v, \bar{v})$ is the Gram form of the Hadamard square $B^{\circ 2} = [B(\xi, \eta)^2]$.

Proof. With $u = e^{i\xi\theta}$, $w = e^{-i\eta\theta}$: $Lu = -\psi(\xi)u$, $Lw = -\psi(\eta)w$, $L(uw) = -\psi(\xi - \eta)uw$, so $\Gamma(u, w) = B(\xi, \eta)e^{i(\xi-\eta)\theta}$. Then $L\Gamma(u, w) = -\psi(\xi - \eta)B e^{\cdot}$, $\Gamma(Lu, w) = -\psi(\xi)B e^{\cdot}$, $\Gamma(u, Lw) = -\psi(\eta)B e^{\cdot}$, whence $\Gamma_2 = \frac{1}{2}B[\psi(\xi) + \psi(\eta) - \psi(\xi - \eta)]e^{\cdot} = B^2 e^{\cdot}$. $\square$

Theorem 5.3 ($CD(0, \infty)$ for the prime kernel; unconditional). *If all jump weights $c_j \geq 0$ (the von Mangoldt case, ζ), then B is a positive-definite kernel, hence $B^{\circ 2}$ is positive-definite (Schur product theorem), hence $\Gamma_2 \geq 0$: the prime Dirichlet generator satisfies $CD(0, \infty)$. The Davenport–Heilbronn kernel (c_j twisted by a non-principal character) is not positive-definite and Γ_2 acquires a negative part.*

Proof. Using $\cos a(\xi - \eta) = \cos a\xi \cos a\eta + \sin a\xi \sin a\eta$,

$$\psi(\xi) + \psi(\eta) - \psi(\xi - \eta) = \sum_j 2c_j [(1 - \cos a_j \xi)(1 - \cos a_j \eta) + \sin a_j \xi \sin a_j \eta],$$

so $B = \sum_j c_j (g_j \otimes g_j + h_j \otimes h_j)$ with $g_j(\xi) = 1 - \cos a_j \xi$, $h_j(\xi) = \sin a_j \xi$ — a sum, with coefficients $c_j \geq 0$, of rank-one positive-definite kernels. Hence $B \succeq 0$, and by Schur $B^{\circ 2} \succeq 0$; Theorem 5.2 gives $\Gamma_2 \geq 0$. The twisted case has some $c_j < 0$, breaking positive-definiteness of B . $\square$

Result (curvature positivity = Euler positivity). Theorem 5.3 identifies the curvature positivity of ζ ’s prime kernel with the positivity of the von Mangoldt weights, i.e. with the Euler product ($\log = \Lambda * 1$ with nonnegative coefficients). This is a local, algebraic place where multiplicativity enters; the falsifier breaks it. The archimedean Laplacian ($\psi = \xi^2$, $B_\phi(\xi, \eta) = \xi\eta$) is also positive-definite, and the combined cross term $B_\phi \circ B_{\text{prime}}$ is positive-definite by Schur, so $\Gamma_2 \geq 0$ persists for the full generator.

5.3. The honest limit of curvature: marginality. $CD(0, \infty)$ is the flat case and gives no spectral gap: $\Gamma_2 \geq \rho \Gamma$ amounts to $\psi(\xi) \geq \rho$, which fails near $\xi = 0$ since $\psi(0) = 0$. The genuine gap is the Dirichlet ground state $(\pi/L)^2$ of the confinement (Lemma 2.3.F), but this is the gap of the *Doob-reduced* G_λ , i.e. $\varepsilon_1 - \varepsilon_0$, not ε_0 itself.

Result (shape vs. sign — the crystallization). Every structure on the route — curvature, de Branges colligation, Kreĭn–Langer index, Hodge-index surface — is built from $G_\lambda = A - \varepsilon_0$ and is therefore *blind to the additive constant* ε_0 . It proves the *shape* positivity unconditionally (the $(k+1)^2$ ladder, $CD(0, \infty)$, the internal gap $(\pi/L)^2$). RH is the *sign* $\varepsilon_0 \geq 0$ — the one quantity invisible to all Doob/scale-invariant structure. This explains uniformly why every route in the wider programme terminates at the same wall.

6. THE SIGN AS A SCHRÖDINGER BOUND STATE: THE POLE AS BARRIER

In the geometric coordinate $\theta = \pi y/L \in [0, \pi]$ the Weil form is a Dirichlet Schrödinger operator

$$H_\lambda = -\frac{d^2}{d\theta^2} + W_\lambda(\theta), \quad W_\lambda = W_\lambda^{\text{pole}} + W_\lambda^{\text{arch}} - W_\lambda^{\text{prime}}, \quad \varepsilon_0(\lambda) = \inf \text{spec } H_\lambda,$$

so $\varepsilon_0 \geq 0 \Leftrightarrow H_\lambda$ has no negative bound state. The pole of ζ at $s = 1$ contributes $W^{\text{pole}} = +p_\lambda \geq 0$: a strictly positive, λ -stable *repulsive barrier*. The prime part is a *well*.

Proposition 6.1 (Birman–Schwinger form). $\varepsilon_0(\lambda) \geq 0 \Leftrightarrow \|K_\lambda(0)\| \leq 1$, where $K_\lambda(0) = G_{\text{Dir}}^{1/2} W_\lambda^- G_{\text{Dir}}^{1/2}$ is the Birman–Schwinger operator, G_{Dir} the Dirichlet Green’s function (gap 1), and W_λ^- the negative part of W_λ (the prime well minus the pole barrier). The Perron–Frobenius u_0 is the threshold resonance: it is the eigenvector of $K_\lambda(0)$ at eigenvalue 1, pinned there by the pole.

Remark 6.2 (a self-correction). A Lieb–Thirring/trace bound $\text{Tr } K_\lambda < 1$ is too lossy at marginality: $\text{Tr } K_\lambda \geq \|K_\lambda\|$, and at the edge $\|K_\lambda\| = 1$ while $\text{Tr } K_\lambda > 1$. The sharp tool is the Birman–Schwinger principle (the top eigenvalue), not the trace.

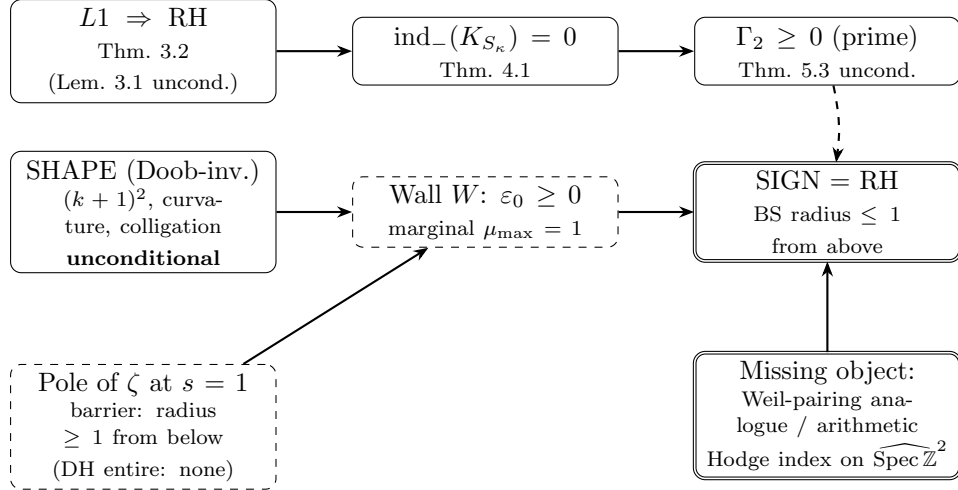
Open target (the pole-pinned spectral radius). Since $G_{\text{Dir}} > 0$ and $W^- \geq 0$, $K_\lambda(0)$ is positivity-improving, so by Perron–Frobenius its top eigenvector is the positive u_0 . Thus

$\text{RH} \iff$ the spectral radius of the pole-barriered von Mangoldt $K_\lambda(0)$ equals 1 uniformly in λ .

The pole pins the radius ≥ 1 from below (u_0 resonance); the DH falsifier, being entire, has *no pole*, hence no barrier, hence radius > 1 and $\varepsilon_0 < 0$ — the cleanest statement of why the falsifier cannot be positive. **What pins the radius ≤ 1 from above is RH and is the missing object.**

7. THE MAP AND THE MISSING OBJECT

7.1. The route as a map.



Solid boxes are unconditional; dashed boxes meet at the wall; double-ruled boxes are the open content. The pole pins the Birman–Schwinger radius from below; the missing object would pin it from above.

7.2. What must be created. The upper pinning is, in the function-field model, the *Weil pairing*: Frobenius acts on H^1 of a curve $C/\mathbb{F}_q$ as an isometry (up to q) of a positive intersection form, so its eigenvalues lie on $|z| = \sqrt{q}$ (the curve RH). The $\widehat{\text{Spec } \mathbb{Z}}$ analogue requires an arithmetic intersection theory on the square of the Connes–Consani arithmetic site $\widehat{\text{Spec } \mathbb{Z}}$ with Frobenius correspondences $[F_t]$, a diagonal $[\Delta]$, and a Hodge-index sign, in which the realization $f \mapsto Z_f$ satisfies $(Z_f \cdot Z_f) = W(f)$ (formally, by the trace formula) and Abboud’s arithmetic Hodge index $(D \cdot D) \leq 0$ for primitive D applies. The degree condition $(Z_f \cdot H) = 0$ is exactly $\mu_{\max} = 1$.

Open target (the single remaining theorem, M3a). Construct $\widehat{\text{CH}}^1(\widehat{\text{Spec } \mathbb{Z}^2})$ with a symmetric pairing for which the scaling correspondences $[F_t]$ and the diagonal are classes, $(Z_f \cdot Z_f) = W(f)$, and the Hodge-index inequality holds; then RH. This is the Weil-pairing analogue over $\text{Spec } \mathbb{Z}$; it does not presently exist and must be *created*.

8. SYNTHESIS

The route is assembled rigorously around a single arithmetic-geometric obstruction:

$$\text{RH} \iff \underbrace{[\text{model space, Lax–Phillips, ind}_-, \Gamma_2 \geq 0, \mu_{\max} = 1, \text{Abboud}]}_{\text{rigorous / assembled}} \iff \underbrace{\text{M3a}}_{\text{open: the upper pinning}}.$$

Three facts make this the honest endpoint. First, $L1 \Rightarrow \text{RH}$ (Theorem 3.2): the boundedness is not a lemma below RH, it is RH; the Laplace–pole obstruction is exactly the off-line poles, so no analytic averaging can reach it (the Cesàro and monotone-continuation routes are RH-equivalent).

Second, $\Gamma_2 \geq 0$ (Theorem 5.3): the curvature/shape positivity is unconditional and rooted in $\Lambda \geq 0$; the falsifier fails it. Third, the wall is the *sign* $\varepsilon_0 \geq 0$, the Birman–Schwinger spectral radius pinned at 1 — from below by the pole (which DH lacks), from above by the missing Weil-pairing analogue.

The geometry proves the shape; RH is the sign; the sign is the upper pinning of the spectral radius; the upper pinning is a piece of arithmetic geometry over $\text{Spec } \mathbb{Z}$ that does not yet exist and must be created.

9. OPEN PROBLEMS

- O1. M3a (the upper pinning).** Construct the arithmetic intersection pairing on $\widehat{\text{Spec } \mathbb{Z}}^2$ with Frobenius correspondences and the Hodge-index sign. Candidate input: the two-variable kernel $\xi(\frac{1}{2} + iz)\xi(\frac{1}{2} + iw)$ and the arithmetic-site Frobenius.
- O2. Sharp curvature.** Is there a curvature-dimension refinement $CD(0, N)$ (finite N , from the archimedean diffusion) that survives the Doob reduction and reaches ε_0 , or is the Doob-invariance obstruction absolute?
- O3. Positivity-improving with pinned radius.** Identify a structural reason the pole-barriered von Mangoldt Birman–Schwinger kernel has spectral radius exactly 1, analogous to the Frobenius $\sqrt{q}$ -isometry; this is the operator form of O1.
- O4. Cyclicity for $L1$.** Prove the cyclicity/density of the near-radical windows needed for the Jacobi-coordinate phrasing of Theorem 3.2 (the forward direction is already coordinate-free).

10. CONCLUSION

We carried the de Branges/colligation route to its rigorous limit. The boundedness $L1$ is proved equivalent to RH with an unconditional Paley–Wiener lemma; the correct invariant is the Kreĭn–Langer index; the prime generator has unconditionally nonnegative Bakry–Émery curvature by a Schur-product mechanism rooted in $\Lambda \geq 0$; and the wall is realized as a pole-pinned Birman–Schwinger spectral radius, with the pole providing the lower pinning that the Davenport–Heilbronn falsifier lacks. The remaining content is one object — the Weil-pairing analogue over $\text{Spec } \mathbb{Z}$ — which provides the upper pinning and which does not yet exist. Crossing the wall is, precisely, creating it. Nothing here proves RH; the boundary is mapped exactly, and the next step is an act of construction in arithmetic geometry.

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